Polar Coordinates and Conversions

ANSWER KEY

Section 1: Plotting with Positive and Negative Radii

1 How is a point with negative radius $(-r, \theta)$ plotted in polar coordinates?

a. **Plot along the opposite ray directly across the pole (angle $\theta + \pi$)** ✓

b. Plot in the negative y direction

c. Reflect across the y-axis

d. The point is undefined

A negative radius $-r$ means moving a distance r backward along the line passing through angle θ, which lands at $(r, \theta + \pi)$.

2 Which polar coordinate represents the same point as $(3, \pi/3)$?

a. $(-3, 4\pi/3)$ ✓

b. $(-3, 2\pi/3)$

c. $(3, 4\pi/3)$

d. $(-3, -\pi/3)$

Adding π to the angle and negating the radius gives $(-3, \pi/3 + \pi) = (-3, 4\pi/3)$.

3 In which quadrant does the polar point $(4, 7\pi/6)$ lie?

a. **Quadrant III** ✓

b. Quadrant II

c. Quadrant IV

d. Quadrant I

Angle $7\pi/6 = 210^\circ$ lies in Quadrant III. With positive radius $r = 4$, the point is in Quadrant III.

Section 2: Converting Polar to Cartesian Coordinates

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4 State the formulas to convert polar coordinates (r, θ) to Cartesian coordinates (x, y).

a. $x = r\cos\theta, y = r\sin\theta$ ✓

b. $x = r\sin\theta, y = r\cos\theta$

c. $x = r^2\cos\theta, y = r^2\sin\theta$

d. $x = \frac{\cos\theta}{r}, y = \frac{\sin\theta}{r}$

By standard right-triangle trigonometry, $x = r\cos\theta$ and $y = r\sin\theta$.

5 Convert the polar point $(6, 2\pi/3)$ to exact Cartesian coordinates.

a. $(-3, 3\sqrt{3})$ ✓

b. $(3, -3\sqrt{3})$

c. $(-3\sqrt{3}, 3)$

d. $(-3, -3\sqrt{3})$

$x = 6\cos(2\pi/3) = 6(-1/2) = -3$. $y = 6\sin(2\pi/3) = 6(\sqrt{3}/2) = 3\sqrt{3}$.

6 Convert $(4, -\pi/4)$ to Cartesian coordinates.

a. $(2\sqrt{2}, -2\sqrt{2})$ ✓

b. $(-2\sqrt{2}, 2\sqrt{2})$

c. $(2\sqrt{2}, 2\sqrt{2})$

d. $(4, -4)$

$x = 4\cos(-\pi/4) = 4(\sqrt{2}/2) = 2\sqrt{2}$. $y = 4\sin(-\pi/4) = 4(-\sqrt{2}/2) = -2\sqrt{2}$.

Section 3: Converting Cartesian to Polar Coordinates

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7 Convert the Cartesian point $(-2, 2)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $(2\sqrt{2}, 3\pi/4)$ ✓

b. $(2\sqrt{2}, 5\pi/4)$

c. $(4, 3\pi/4)$

d. $(2\sqrt{2}, \pi/4)$

$r = \sqrt{(-2)^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$. In Quadrant II, $\theta = 3\pi/4$.

8 Convert $(0, -5)$ to polar coordinates.

a. $(5, 3\pi/2)$ ✓

b. $(5, \pi/2)$

c. $(-5, 0)$

d. $(5, \pi)$

Distance $r = 5$. The point lies on the negative y-axis, which is angle $\theta = 3\pi/2$.

9 Convert $(3, -3\sqrt{3})$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $(6, 5\pi/3)$ ✓

b. $(6, 4\pi/3)$

c. $(6, 2\pi/3)$

d. $(12, 5\pi/3)$

$r = \sqrt{9 + 27} = \sqrt{36} = 6$. In Quadrant IV with $\tan\theta = -\sqrt{3}$, $\theta = 2\pi - \pi/3 = 5\pi/3$.

Section 4: Converting Equations Between Polar and Cartesian

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10 Convert the polar equation $r = 4\cos\theta$ to Cartesian form.

a. $(x-2)^2 + y^2 = 4$ ✓

b. $x^2 + y^2 = 4$

c. $x^2 + (y-2)^2 = 4$

d. $x = 4$

Multiply by r: $r^2 = 4r\cos\theta \Rightarrow x^2 + y^2 = 4x \Rightarrow (x-2)^2 + y^2 = 4$.

11 Convert $r = 5\csc\theta$ to Cartesian form.

a. $y = 5$ ✓

b. $x = 5$

c. $x^2 + y^2 = 25$

d. $y = 5x$

$r = \frac{5}{\sin\theta} \Rightarrow r\sin\theta = 5 \Rightarrow y = 5$.

12 Convert the line $y = 3x$ to a polar equation.

a. $\theta = \arctan(3)$ ✓

b. $r = 3$

c. $r = 3\cos\theta$

d. $\theta = 3$

Substitute $x = r\cos\theta, y = r\sin\theta$: $r\sin\theta = 3r\cos\theta \Rightarrow \tan\theta = 3 \Rightarrow \theta = \arctan(3)$.

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